



US 20250277733A1

(19) **United States**(12) **Patent Application Publication**
HASEGAWA et al.(10) **Pub. No.: US 2025/0277733 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ACCELERATED DETERIORATION TEST APPARATUS, ACCELERATED-DETERIORATION-TEST ANALYSIS SYSTEM, AND ACCELERATED DETERIORATION TEST METHOD**(71) Applicant: **SHIMADZU CORPORATION,**
Kyoto-shi (JP)(72) Inventors: **Yukinori HASEGAWA,** Kyoto-shi (JP);
Kenichi KITAMURA, Kyoto-shi (JP);
Masaru FURUTA, Kyoto-shi (JP);
Ryutaro ODA, Kyoto-shi (JP)(21) Appl. No.: **18/858,979**(22) PCT Filed: **Jun. 12, 2023**(86) PCT No.: **PCT/JP2023/021697**

§ 371 (c)(1),

(2) Date: **Oct. 22, 2024**(30) **Foreign Application Priority Data**

Jun. 27, 2022 (JP) 2022-102625

Publication Classification(51) **Int. Cl.**
G01N 17/00 (2006.01)(52) **U.S. Cl.**
CPC **G01N 17/004** (2013.01)(57) **ABSTRACT**

An accelerated deterioration test apparatus (101) according to this invention includes a container (10) configured to be able to accommodate a sample (200) as a subject of an accelerated deterioration test for accelerating deterioration of the sample (200) as the test subject by using a deterioration factor for deteriorating the sample; and an introducer/collector (120) connected to the container (10) to introduce a gas or a liquid as the deterioration factor into the container (10) and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample (200).

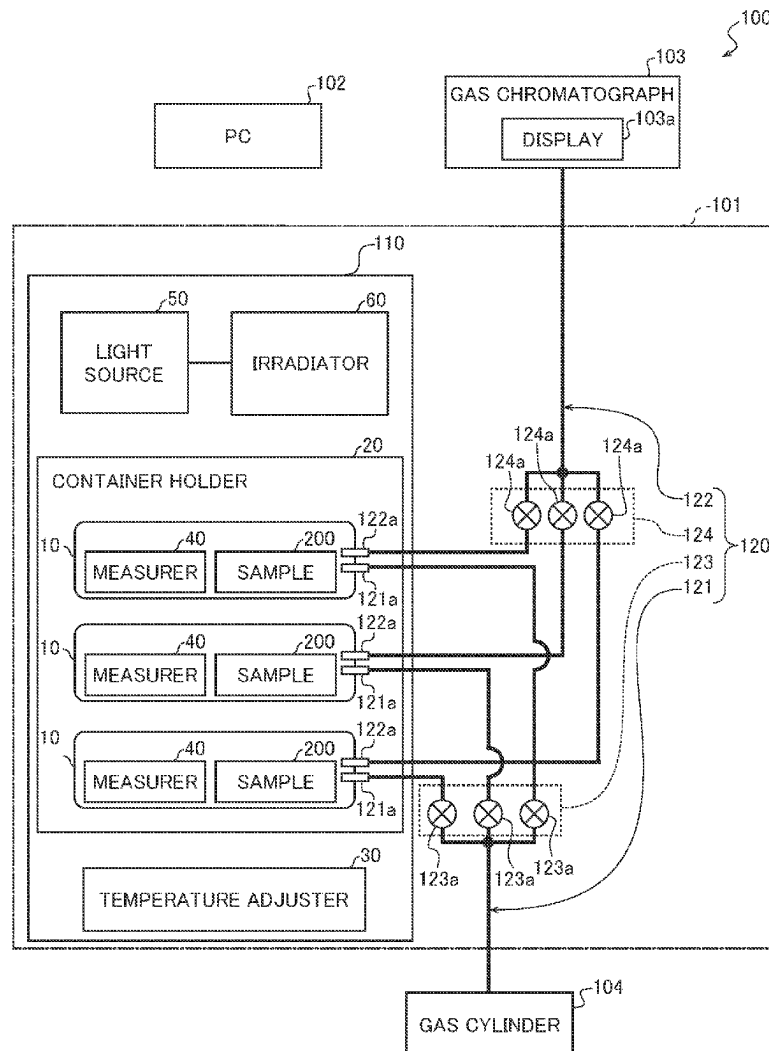


FIG. 1

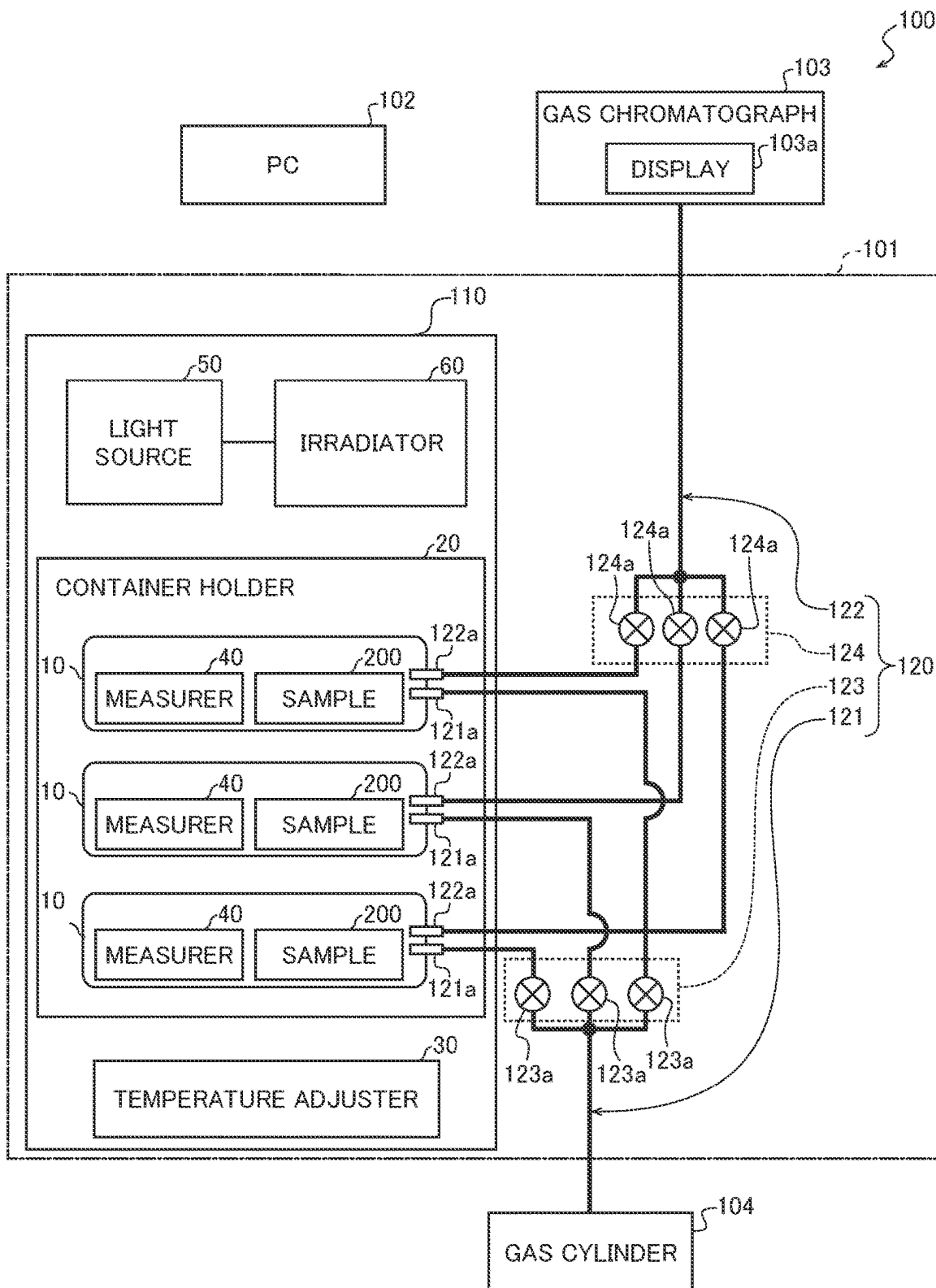


FIG.2

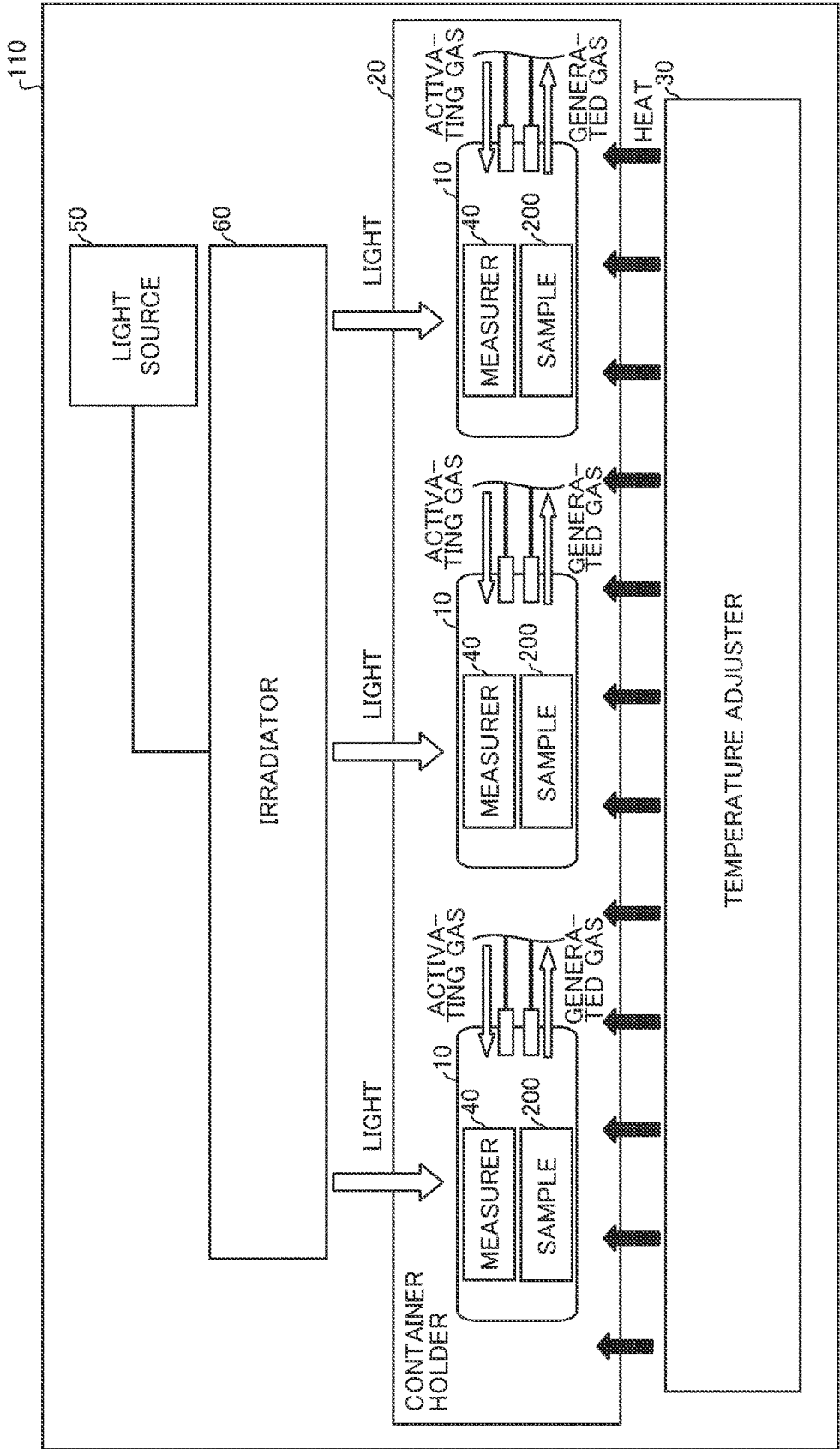


FIG.3

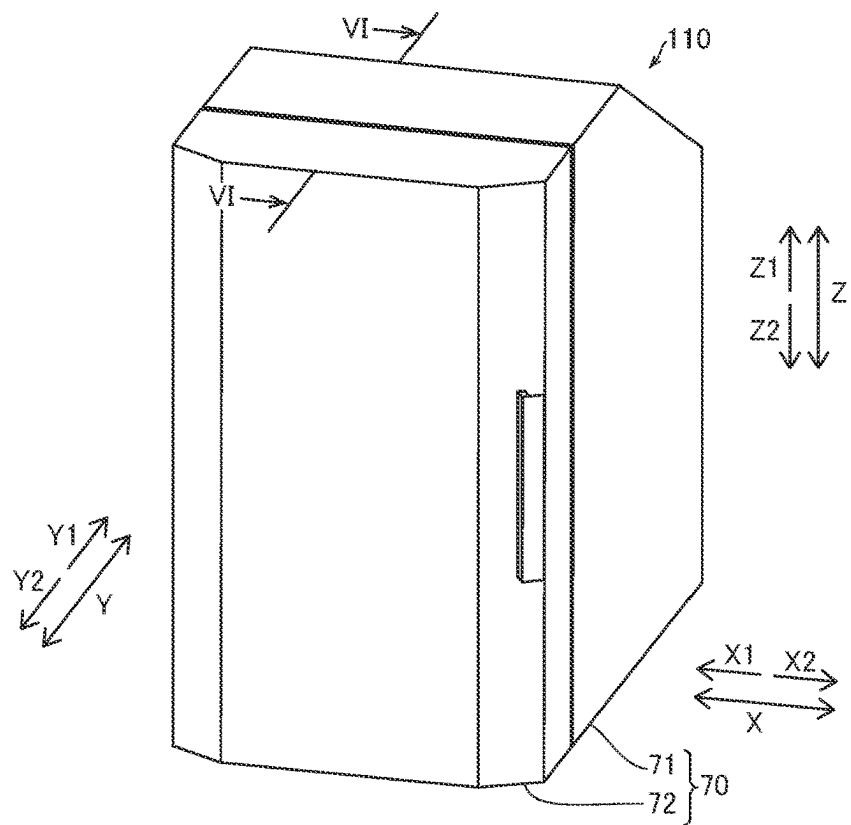


FIG.4

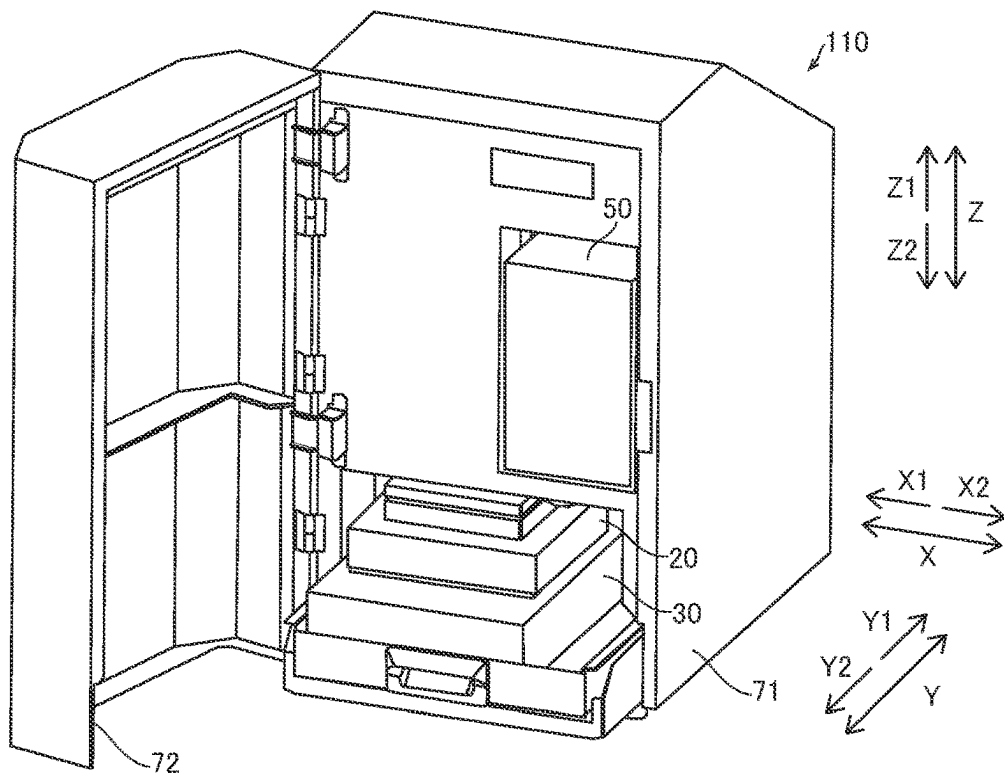


FIG.5

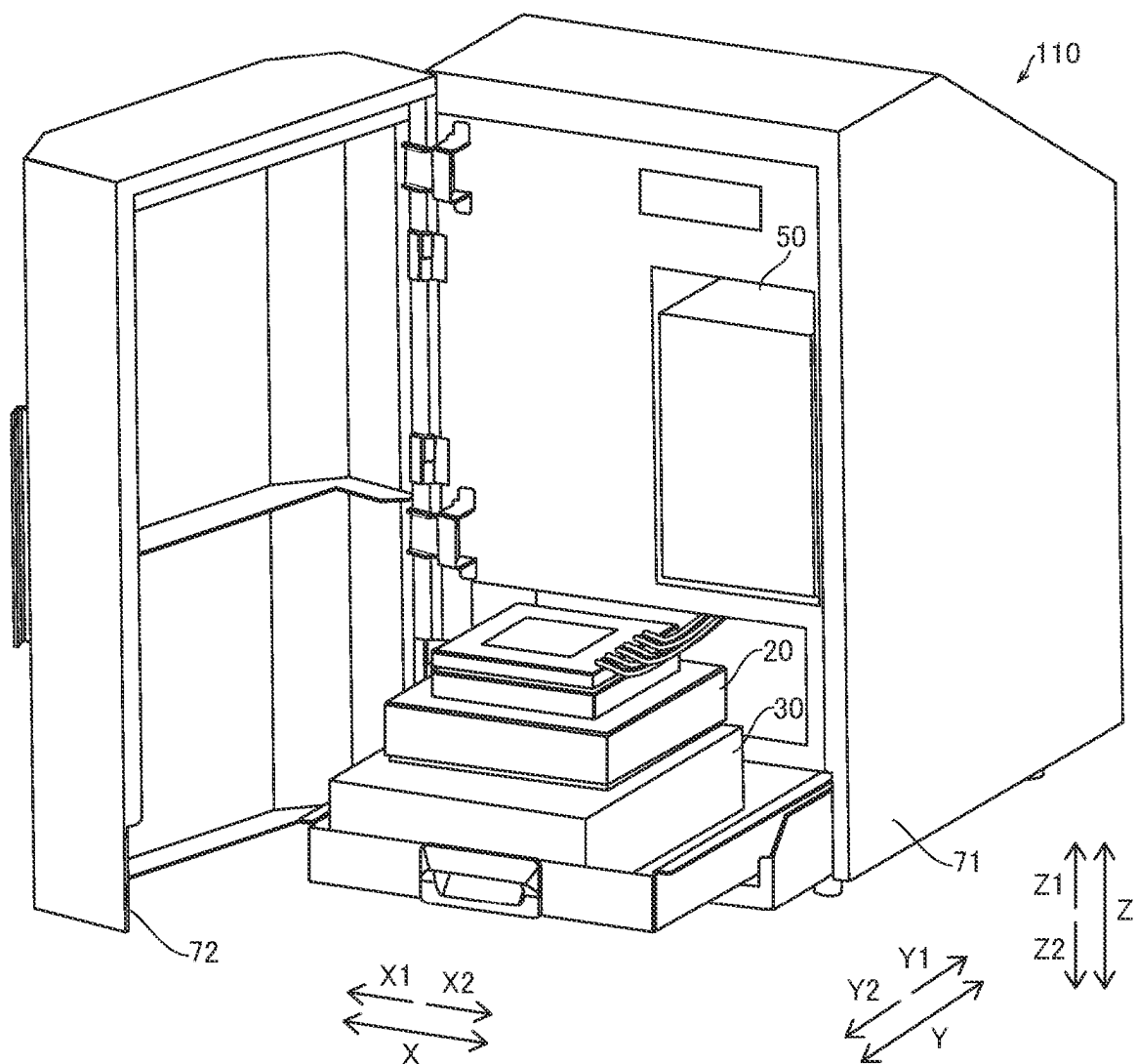


FIG. 6

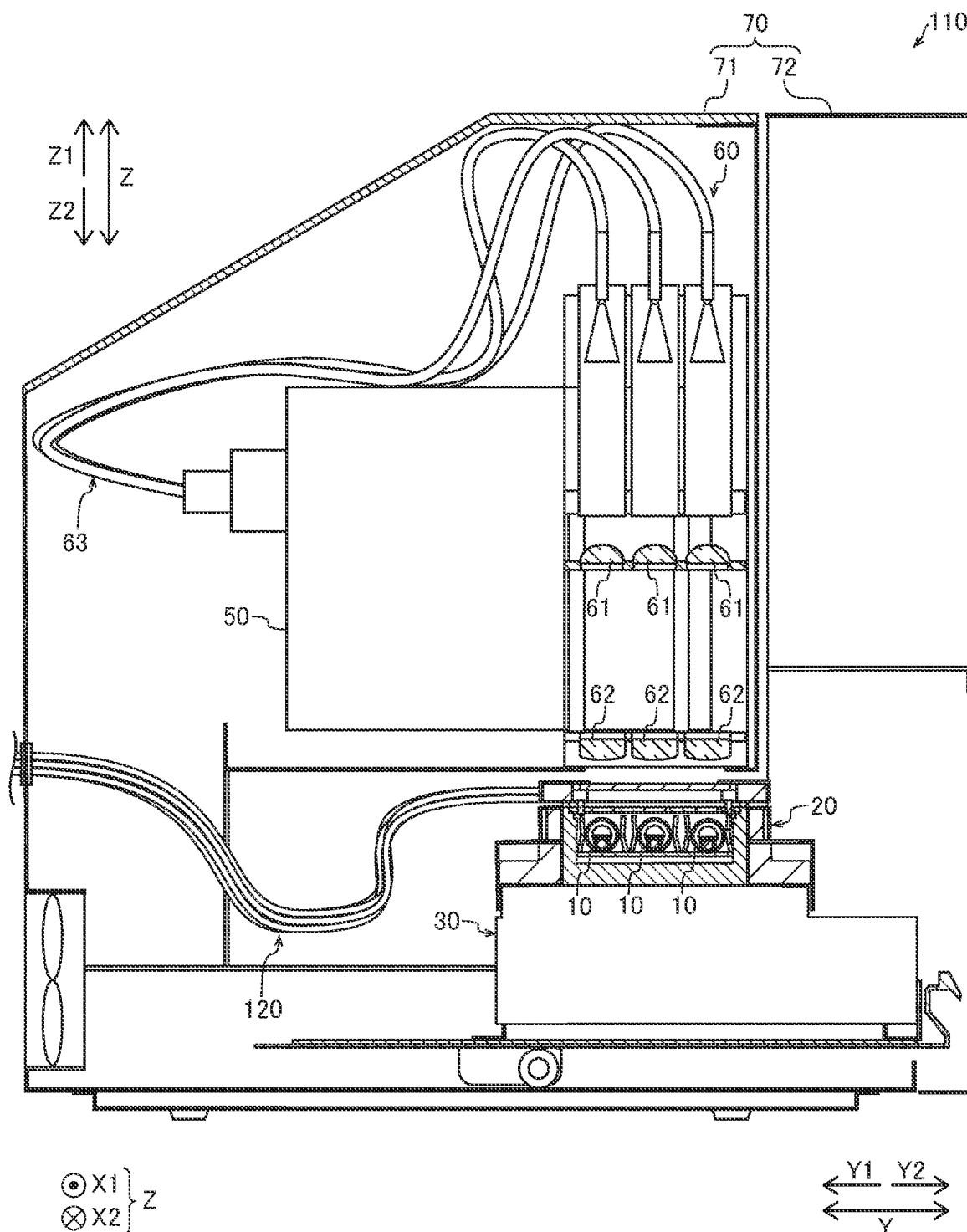


FIG. 7

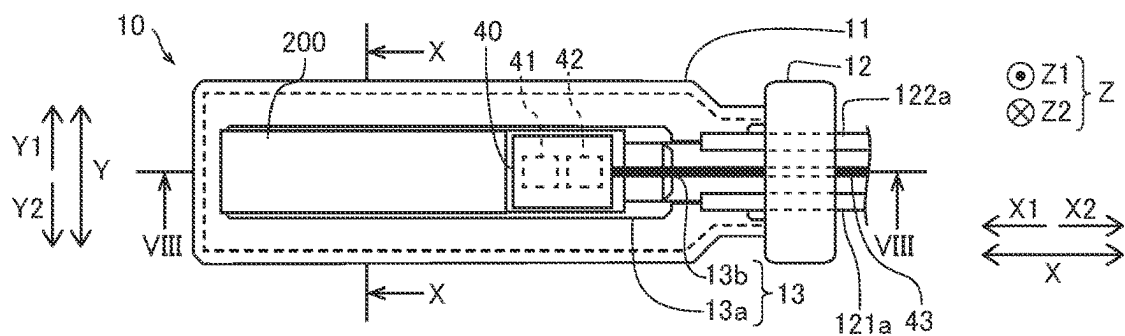


FIG. 8

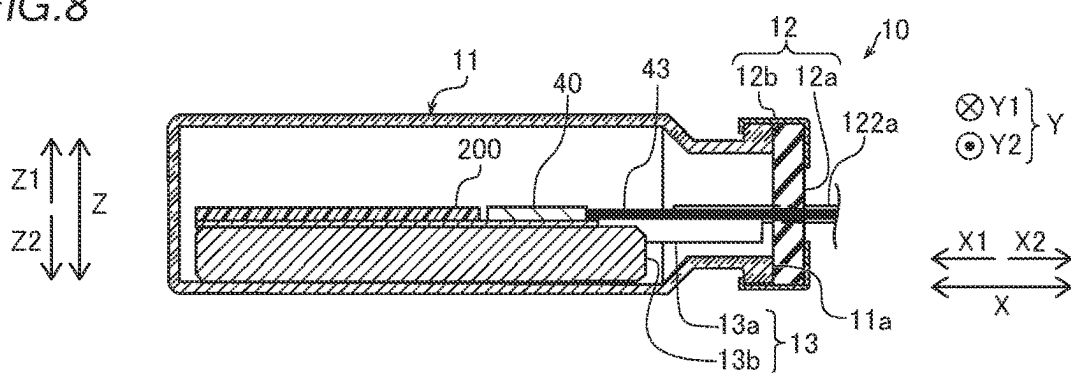


FIG. 9

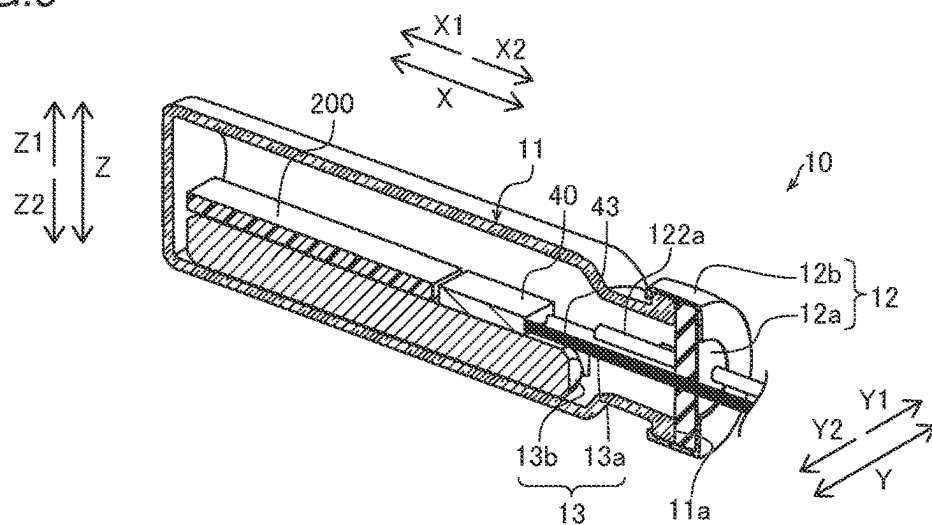


FIG. 10

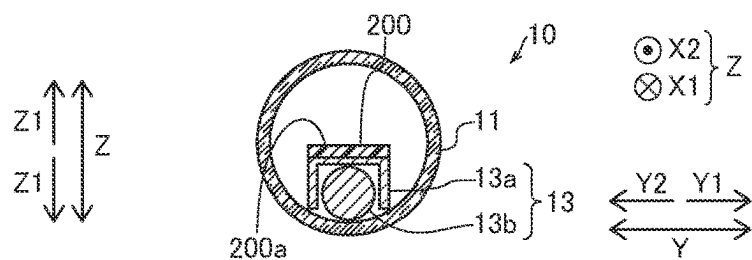


FIG. 11

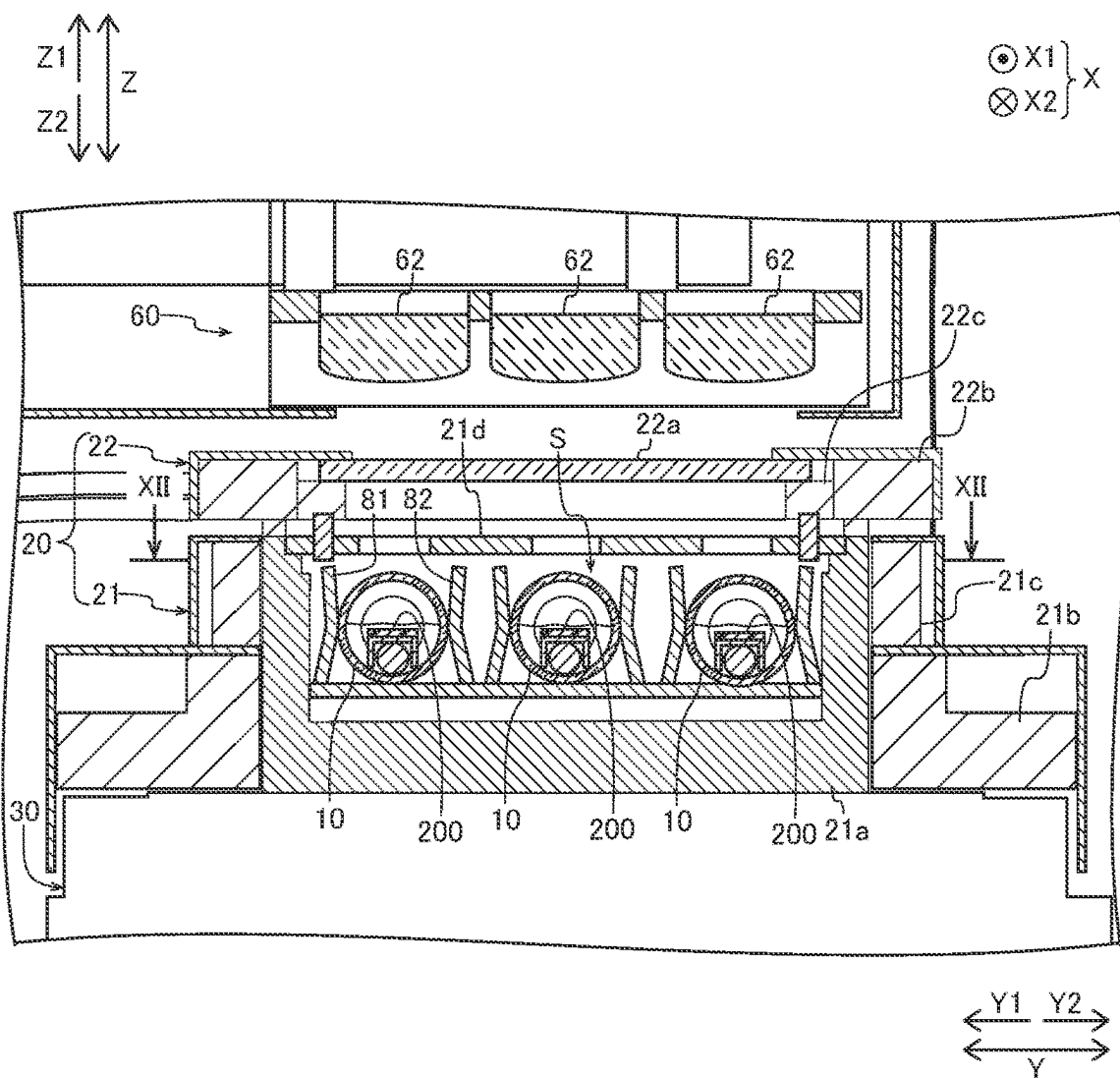


FIG. 12

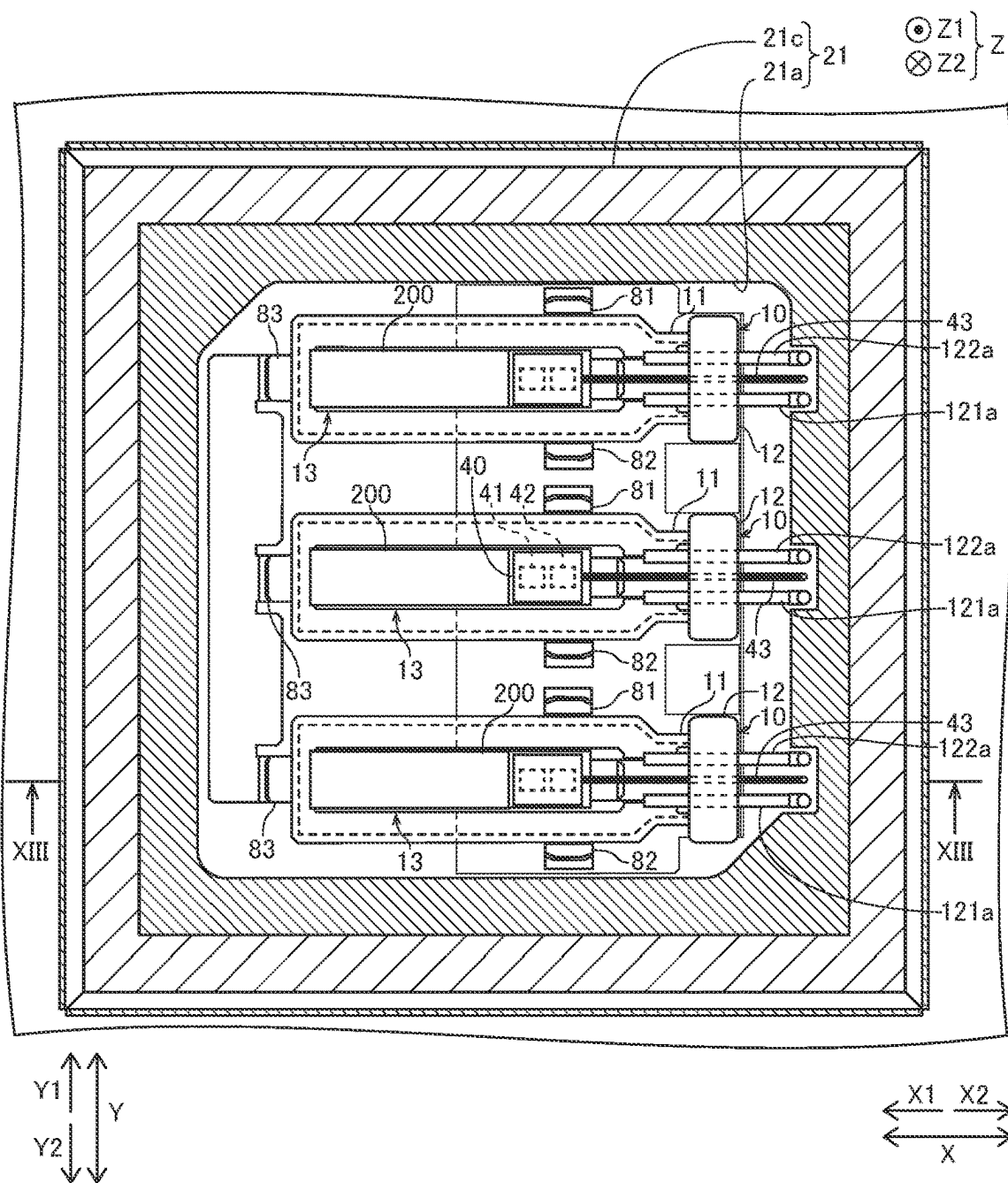


FIG. 13

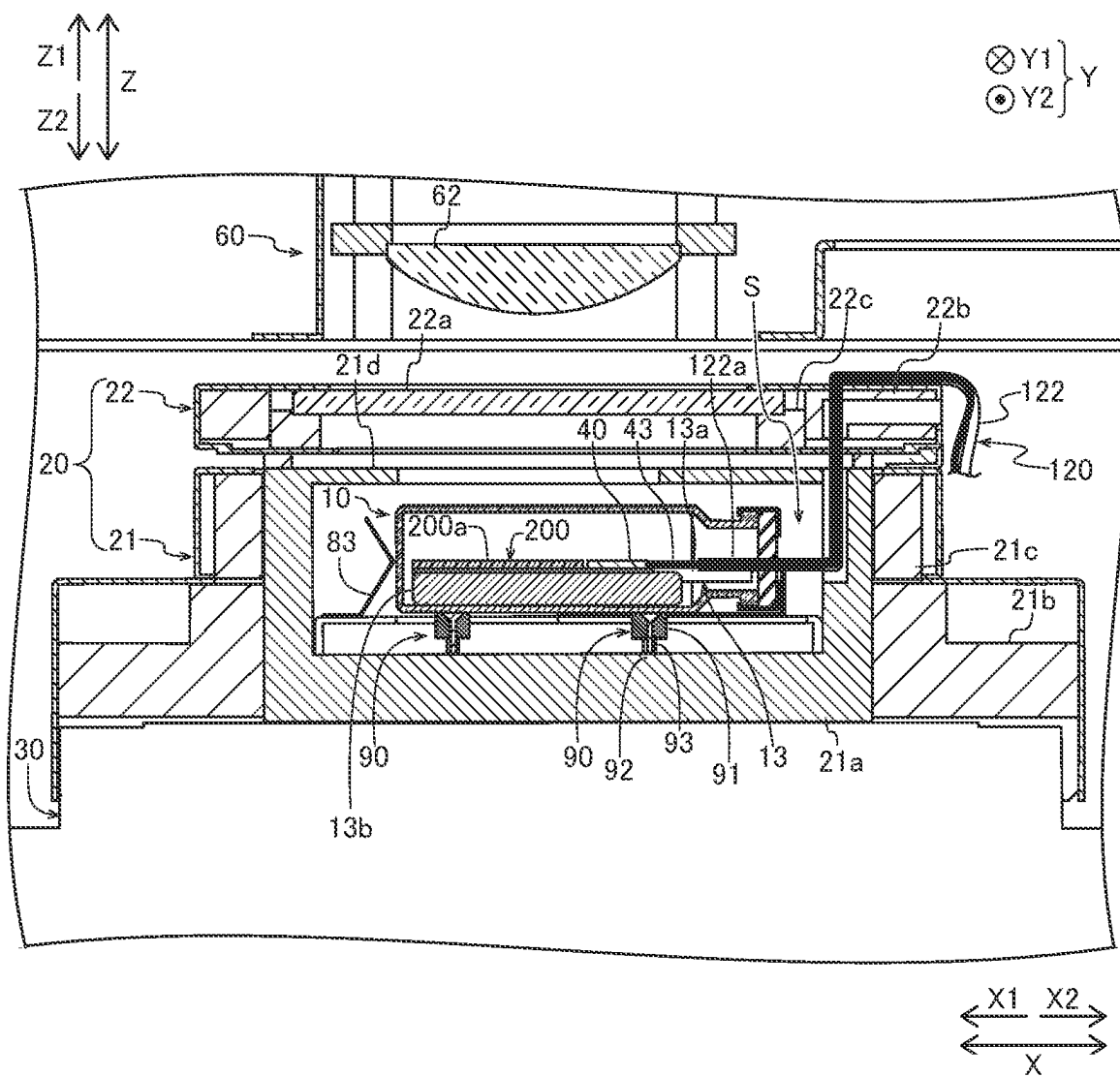


FIG. 14

ACCELERATED DETERIORATION TEST AND
ANALYSIS PROCESSING OF GENERATED GASES

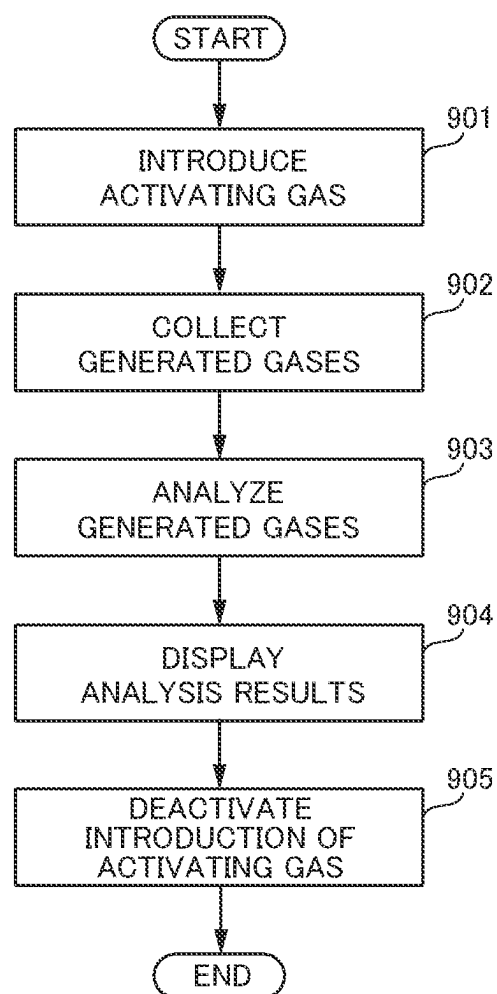
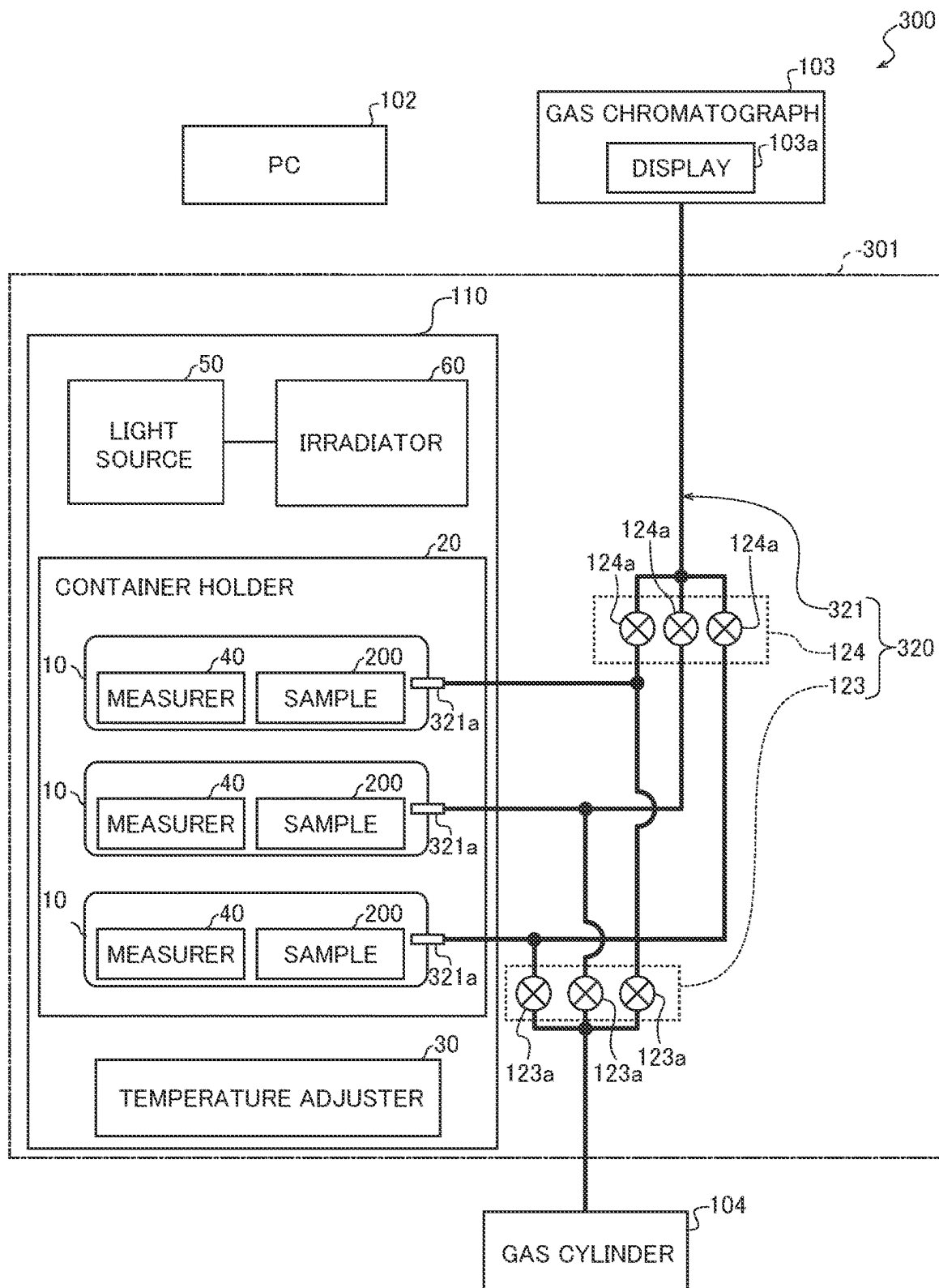


FIG. 15



**ACCELERATED DETERIORATION TEST
APPARATUS,
ACCELERATED-DETERIORATION-TEST
ANALYSIS SYSTEM, AND ACCELERATED
DETERIORATION TEST METHOD**

TECHNICAL FIELD

[0001] The present invention relates to an accelerated deterioration test apparatus, an accelerated-deterioration-test analysis system, and an accelerated deterioration test method.

BACKGROUND ART

[0002] Combination cycle testers are known for evaluating corrosion resistances of various types of materials by applying tests to various types of materials under a plurality of combined types of test conditions (environmental tests) in the art. Such an apparatus is disclosed in Japanese Patent Laid-Open Publication No. JP 2018-132385, for example.

[0003] In the combination cycle tester in the aforementioned Japanese Patent Laid-Open Publication No. JP 2018-132385, an accelerated deterioration test for accelerating deterioration of various types of materials is conducted in a test chamber while adjusting deterioration factors (deterioration loads) such as temperature and humidity. This combined cycle tester introduces outside air to maintain temperature and humidity conditions in the test chamber. Also, the accelerated deterioration test for accelerating deterioration of a sample is conducted by spraying a liquid such as salt water (corrosive liquid) while maintaining conditions for acceleration of deterioration of the sample by introducing outside air (air) and circulating the outside air through the test chamber.

PRIOR ART

Patent Document

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. JP 2018-132385

SUMMARY OF THE INVENTION

Problems to Be Solved by the Invention

[0005] However, in the combination cycle tester stated in the aforementioned Japanese Patent Laid-Open Publication No. JP 2018-132385, the air circulated through the test chamber and the corrosive liquid sprayed onto the sample during the accelerated deterioration test are discharged to the outside of the test chamber. Accordingly, since products such as gases, liquids, or precipitates (solids) produced by the deterioration of the sample caused by the corrosion liquid are also discharged to the outside of the test chamber, a deterioration state of the sample during the accelerated deterioration test cannot be acquired based on the products, which are produced by the deterioration of the sample during the accelerated deterioration test. For this reason, the deterioration state of the sample during the accelerated deterioration test cannot be acquired while maintaining conditions for acceleration of deterioration of the sample.

[0006] The present invention is intended to solve the above problem, and one object of the present invention is to provide an accelerated deterioration test apparatus, an accelerated-deterioration-test analysis system, and an accelerated

deterioration test method capable of acquiring a deterioration state of a sample during an accelerated deterioration test while maintaining conditions for acceleration of deterioration of the sample.

Means for Solving the Problems

[0007] In an accelerated deterioration test apparatus according to a first aspect of the present invention includes a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample; and an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample.

[0008] In an accelerated-deterioration-test analysis system according to a second aspect of the present invention includes a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample; an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample; and an analyzer for analyzing the product, which is collected by using the introducer/collector.

[0009] In an accelerated deterioration test method according to a third aspect of the present invention includes an accelerated deterioration test step of conducting an accelerated deterioration test for accelerating deterioration of a sample as a test subject by introducing a gas or a liquid as a deterioration factor for deteriorating the sample by using an introducer/collector connected to a container for accommodating the sample to introduce the gas or the liquid as the deterioration factor into the container and to collect a product that appears with the deterioration of the sample from the container with the sample being accommodated in a container; and a collection step of collecting the product, which appears with deterioration of the sample, from the container by using the introducer/collector during the accelerated deterioration test.

Effect of the Invention

[0010] In the accelerated deterioration test apparatus according to the first aspect, the accelerated-deterioration-test analysis system according to the second aspect, and the accelerated deterioration test method according to the third aspect of the present invention, an accelerated deterioration test is conducted by introducing a gas or a liquid as a deterioration factor into a container accommodating a sample as a test subject by using an introducer/collector, and a product that appears with deterioration of the sample is collected by using the introducer/collector. Accordingly, since the gas or the liquid as the deterioration factor is introduced into the container by using the introducer/collector during the accelerated deterioration test, it is possible to maintain conditions for acceleration of deterioration of the sample accelerated by the gas or liquid as the deterioration factor. Also, a deterioration state of the sample can be acquired during the accelerated deterioration test by analyzing the product collected by the introducer/collector during

the accelerated deterioration test. For these reasons, the deterioration state of the sample during the accelerated deterioration test can be acquired while maintaining conditions for acceleration of deterioration of the sample. In addition, since the product, which appears with deterioration of the sample, is collected during the accelerated deterioration test, it is possible to prevent the product, which appears with deterioration of the sample, from affecting the accelerated deterioration test.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a schematic diagram entirely showing a configuration of an accelerated-deterioration-test analysis system according to one embodiment of the present invention.

[0012] FIG. 2 is a schematic diagram entirely showing a configuration of a test apparatus main body of an accelerated deterioration test apparatus according to the one embodiment of the present invention.

[0013] FIG. 3 is a perspective view showing an outer appearance of the accelerated deterioration test apparatus main body.

[0014] FIG. 4 is a perspective view showing the accelerated deterioration test apparatus with its door being opened.

[0015] FIG. 5 is a perspective view showing the accelerated deterioration test apparatus main body with its container being pulled out.

[0016] FIG. 6 is a cross-sectional view of the accelerated deterioration test apparatus main body taken along a line VI-VI in FIG. 3.

[0017] FIG. 7 is a view of an outer appearance of a container.

[0018] FIG. 8 is a cross-sectional view of the container taken along a line VIII-VIII in FIG. 7.

[0019] FIG. 9 is a perspective cross-sectional view of the container taken along the line VIII-VIII in FIG. 7.

[0020] FIG. 10 is a cross-sectional view of the container taken along a line X-X in FIG. 7.

[0021] FIG. 11 is a partially enlarged view of a peripheral part of containers each of which is shown in FIG. 6.

[0022] FIG. 12 is a cross-sectional view of the peripheral part taken along a line XII-XII in FIG. 11.

[0023] FIG. 13 is a cross-sectional view of the peripheral part taken along a line XIII-XIII in FIG. 12.

[0024] FIG. 14 is a flowchart showing exemplary accelerated deterioration test and exemplary generated gas analysis processing conducted by the accelerated-deterioration-test analysis system according to the one embodiment of the present invention.

[0025] FIG. 15 is a schematic diagram entirely showing a configuration of an accelerated-deterioration-test analysis system according to a modified embodiment.

MODES FOR CARRYING OUT THE INVENTION

[0026] Embodiments embodying the present invention will be described with reference to the drawings.

[0027] The following description describes a configuration of an accelerated-deterioration-test analysis system 100 and a configuration of an accelerated deterioration test apparatus 101 according to this embodiment with reference to FIGS. 1 to 13.

Configuration of Accelerated Deterioration Test Apparatus

[0028] As shown in FIG. 1, the accelerated-deterioration-test analysis system 100 includes the accelerated deterioration test apparatus 101, a PC (Personal computer) 102, a gas chromatograph 103, and a gas cylinder 104. The PC 102 and the gas chromatograph 103 are an example of a “controller” and an example of an “analyzer, respectively, in the claims. Also, the gas cylinder 104 is an example of a “deterioration factor supply source” in the claims.

[0029] The accelerated deterioration test apparatus 101 is an apparatus for conducting an accelerated deterioration test (accelerated aging test) to accelerate deterioration of samples 200. The accelerated deterioration test apparatus 101 is mainly used for material discovery and in prototyping stages in material development. The samples 200 as test subjects to be tested by the accelerated deterioration test apparatus 101 include raw materials (resins, fibers, metals and wood, and the like), paints, components for transportation machines, oil for food, packaging for food, and food.

[0030] The accelerated deterioration test apparatus 101 includes a test apparatus main body 110 and an introducer/collector 120. As shown in FIG. 1, the test apparatus main body 110 includes containers 10 that can accommodate the samples 200 as the subjects for the accelerated deterioration test, and a container holder 20 that holds the containers 10. Also, the test apparatus main body 110 is relatively small to be installed in a laboratory.

[0031] The test apparatus main body 110 includes a temperature adjuster 30 for adjusting temperature as a deterioration factor, and a measurer 40 for measuring the temperature in the container 10. Also, the test apparatus main body 110 includes a light source 50 for emitting light as a deterioration factor, and an irradiator 60 for irradiating the samples with the light as a deterioration factor emitted from the light source 50.

[0032] In this embodiment, the light source 50 is configured to irradiate the samples with light including ultraviolet rays (UV light). The light source 50 includes UV light source for irradiating the samples with ultraviolet rays, for example. The light source 50 may include a xenon light source including a xenon lamp. That is, the light source 50 may be a light source for emitting light that includes a wavelength range other than ultraviolet rays. Also, the light source 50 may be a light source for emitting infrared light.

[0033] The PC 102 is configured to control sending of products that appear with the deterioration of the samples 200 to the gas chromatograph 103 through the introducer/collector 120. In this embodiment, the PC 102 controls sending of gases appearing with the deterioration of the samples 200 (generated gases) to the gas chromatograph 103 through the introducer/collector 120. A keyboard, a computer mouse, a display and the like (not shown) are connected to the PC 102.

[0034] The gas chromatograph 103 is apparatus for analyzing the products collected by using the introducer/collector 120. The gas chromatograph 103 is configured to analyze the products, which are collected from the containers 10 during the accelerated deterioration test by using the introducer/collector 120, during the accelerated deterioration test. In this embodiment, gas chromatograph 103 analyzes the gases appearing with the deterioration of the samples 200 (generated gases).

[0035] In the accelerated-deterioration-test analysis system 100 according to this embodiment, the gas chromatograph 103 includes a display 103a. The display 103a includes a liquid crystal display or an organic EL display. The display 103a may be a touch panel display including a touch panel. In this embodiment, as described above, the generated gases as the products appearing with the deterioration of the samples 200 are analyzed by the gas chromatograph 103. Specifically, the gas chromatograph 103 analyzes the generated gases, which appear with the deterioration of the samples 200 and are sent to the gas chromatograph 103 through the introducer/collector 120. The analysis results of the generated gases analyzed by the gas chromatograph 103 are displayed on the display 103a.

[0036] The gas cylinder 104 is filled with an activating gas (gas) as a deterioration factor. The gas cylinder 104 supplies the activating gas into the container 10 through the introducer/collector 120. The activating gas introduced into container 10 includes oxygen, an acid gas, an alkaline gas, or the like, for example. In a case in which oxygen is used as the activating gas, an oxidation resistance test (accelerated deterioration test) for the samples 200 can be conducted. In a case in which the acid gas is used as the activating gas, an acid resistance test (accelerated deterioration test) for the samples 200 can be conducted. In a case in which the alkaline gas is used as the activating gas, an alkaline resistance test (accelerated deterioration test) for the samples 200 can be conducted.

[0037] That is, in this embodiment, the test apparatus main body 110 is configured to apply the accelerated deterioration test to the samples 200 accommodated in the containers 10 while introducing the activating gas (gas) as the deterioration factor into the containers 10 accommodating the samples 200. In this embodiment, as shown in FIG. 2, the accelerated deterioration test apparatus 101 introduces the activating gas as the deterioration factor into the containers 10 by using the introducer/collector 120 with the sample 200 being accommodated in the container 10, and accelerates the deterioration of the samples 200 while conducting temperature adjustment and irradiation of the samples 200 with light. Also, the accelerated deterioration test apparatus 101 is configured to collect the products appearing with the deterioration of the samples 200 (generated gases) by using the introducer/collector 120. In other words, the test apparatus main body 110 is a combination accelerated deterioration test apparatus that can conduct an accelerated deterioration test while adjusting a plurality of deterioration factors (applying a plurality of deterioration loads to the samples). Accordingly, it is possible to conduct an accelerated deterioration test (accelerated aging test) under conditions closer to more practical exposure environment (actual exposure environment).

[0038] Also, humidity in the container 10 can be adjusted by water contained in the container 10. The humidity in the container 10 can be changed in accordance with an amount of water vapor in the container 10.

[0039] Each container 10 is configured to be able to accommodate the sample 200 as the test subject in the accelerated deterioration test for accelerating deterioration of the sample 200 by using the plurality of deterioration factors (activating gas, temperature, light and humidity) for accelerating deterioration of the sample 200. In the test apparatus main body 110, an inert gas such as nitrogen and argon can be introduced into the container 10 instead of the

activating gas so that the accelerated deterioration test can be conducted by using at least one of temperature, light and liquid as the deterioration factors (deterioration loads).

[0040] In this embodiment, as shown in FIG. 1, the introducer/collector 120 is connected to the containers 10 to introduce the activating gas as the deterioration factor into the containers 10, and to collect the generated gases as the products appearing with the deterioration of the samples 200 during the accelerated deterioration test. The introducer/collector 120 is connected to the gas chromatograph 103 to be able to send the generated gases appearing with the deterioration of the samples 200 to the gas chromatograph 103.

[0041] The introducer/collector 120 includes an introduction path 121 connecting the gas cylinder 104 to the containers 10, and a collection path 122 connecting the containers 10 to the gas chromatograph 103. The introduction path 121 is a pipe connecting the gas cylinder 104 to the containers 10. Also, the collection path 122 is a pipe connecting the containers 10 to the gas chromatograph 103.

[0042] The introducer/collector 120 includes introduction tubes 121a and collection tubes 122a. Specifically, the introduction path 121 and the collection path 122 include the introduction tubes 121a and the collection tubes 122a, respectively. The introduction tubes 121a and the collection tubes 122a are examples of an “introduction/collection tube” in the claims. The introduction tubes 121a and the collection tubes 122a will be described in detail later.

[0043] The introducer/collector 120 includes a switcher 123 for switching selection of, from the plurality of containers 10, the container 10 to which the activating gas as the deterioration factor is introduced. The switcher 123 is an example of a “second switcher” in the claims. The switcher 123 is arranged on the introduction path 121. The switcher 123 includes adjustment valves 123a provided for the plurality of containers 10. In this embodiment, the switcher 123 includes three adjustment valves 123a corresponding to three containers 10. The adjustment valves 123a are provided between introduction tubes 121a and the gas cylinder 104. Flow rates of the activating gas introduced into the containers 10 can be adjusted by adjusting opening degrees of the adjustment valves 123a.

[0044] The introducer/collector 120 includes a switcher 124 for switching selection of, from the plurality of containers 10, the container 10 from which the product, which appears with the deterioration of the sample 200, is collected. Here, the switcher 124 is an example of a “first switcher” in the claims. The switcher 124 is arranged on the collection path 122. The switcher 124 includes adjustment valves 124a provided for the plurality of containers 10. In this embodiment, the switcher 124 includes three adjustment valves 124a corresponding to three containers 10. The adjustment valves 124a are provided between the collection tubes 122a and the gas chromatograph 103. Flow rates of the generated gases sent from the container 10 to the gas chromatograph 103 can be adjusted by adjusting opening degrees of the adjustment valves 124a.

[0045] The PC 102 controls selection of the container 10 to which the activating gas is introduced by using the switcher 123, and selection of the container 10 from which the generated gas is collected by using the switcher 124. Specifically, the PC 102 controls activation of introduction and deactivation of introduction of the activating gas into the plurality of containers 10 by controlling the opening degrees

of the plurality of adjustment valves **123a** in the switcher **123**, and controls the flow rates of the activating gas introduced into the plurality of containers **10**. Also, the PC **102** controls activation of collection and deactivation of collection of gases that include the generated gases from the plurality of containers **10** by controlling the opening degrees of the plurality of adjustment valves **124a** in the switcher **124**, and controls the flow rates of the gases including the generated gases collected from the plurality of containers **10**. The PC **102** controls selection switched by the switcher **123** and the switcher **124** in accordance with timing of activation and deactivation of analysis conducted by the gas chromatograph **103**. In addition, pumps may be provided on the introduction path **121** and the collection path **122**.

Configuration of Test Apparatus Main Body

[0046] As shown in FIG. 3, the test apparatus main body **110** includes a housing **70** including a main body **71** and an openable/closable door **72** attached to the main body **71**. As shown in FIG. 4, the container holder **20**, the temperature adjuster **30**, and the light source **50** are arranged in the main body **71**. Also, as shown in FIG. 5, the test apparatus main body **110** is configured to be able to move (pull out) the container holder **20** and the temperature adjuster **30** to move in a forward direction (Y2 direction) with the door **72** being opened. When the accelerated deterioration test is conducted, the container holder **20** holding the containers **10** accommodating the samples **200** is placed in the main body **71**.

[0047] Also, the container holder **20** is interposed between the temperature adjuster **30** and the irradiator **60** in a vertical direction (Z direction) in the main body **71**, as shown in FIG. 6. Specifically, the light source **50** and the irradiator **60** are arranged on an upper side (Z1 side) of the container holder **20**, and the temperature adjuster **30** is arranged on a lower side (Z2 side) of the container holder **20**. The container holder **20** is configured to be arranged on an upper surface of the temperature adjuster **30** in the test apparatus main body **110**. The temperature adjuster **30** is configured to increase temperature of air surrounding the containers **10**. The temperature adjuster **30** includes a hotplate, for example.

[0048] The PC **102** controls adjustment of the temperature by using the temperature adjuster **30**, and adjustment of light emitted from the light source **50**. For example, a control board is arranged in the housing **70** so that the control board collectively controls adjustment of the temperature by using the temperature adjuster **30**, and adjustment of light emitted from the light source **50** based on commands from the PC **102**.

[0049] Also, the temperature adjuster **30** may be configured to allow a user to make a change of a setting temperature or the like based on operation of the user on an operation unit (not shown). Also, the light source **50** may be configured to allow the user to make a change of intensity of light emitted from the light source or the like based on operation of the user on the operation unit (not shown). In other words, the user may directly control adjustment of the temperature through the temperature adjuster **30**, and adjustment of light emitted from the light source **50** (light with which the samples are irradiated by the irradiator **60**) without using the PC **102**.

[0050] As shown in FIG. 6, the containers **10** include a plurality of (three) containers. The irradiator **60** is config-

ured to divert the light emitted from the light source **50**, and to irradiate the samples **200**, which are accommodated in the plurality of containers **10** held in the container holder **20**, with the light.

[0051] If the intensity of light emitted by the light source **50** is low, the sample **200** that is located away from a center of an irradiation area cannot be irradiated with sufficient intensity light required for deterioration, and as a result speedup (acceleration) of deterioration due to light may vary depending on a position of the sample in the container **10** in the container holder **20**. Contrary to this, in this embodiment, as described above, the irradiator **60** irradiates the samples **200** in the plurality of containers **10** held in the container holder **20** one by one. Accordingly, it is possible to irradiate the samples **200** in the plurality of containers **10** one by one dissimilar to a case in which samples in the plurality of containers **10** are collectively irradiated with light. As a result, even if the intensity of light emitted by the light source **50** is low, it is possible to surely provide a certain amount of intensity of light with which the samples **200** accommodated in the plurality of containers **10** are irradiated independent of the position of the sample in the container **10** in the container holder **20**. Consequently, it is possible to reduce variation of speedup (acceleration) of deterioration due to light depending on the position of the sample in the container **10** in the container holder **20**.

[0052] The irradiator **60** includes lenses **61** and **62**. The number of the lenses **61** or **62** is the same as the number of containers **10** that can be held in the container holder **20**. In this embodiment, the container holder **20** can hold three containers **10**, and three lenses **61** and three lenses **62** are included. Also, the irradiator **60** includes a light guide **63**. The light guide **63** includes optical fibers, and is configured to guide the light emitted from the light source **50**. The light guide **63** has one end connected to the light source **50**, and has three branches on another end. The light emitted from the light source **50** is divided into three light branches by the light guide **63** so that each of the light branches enters corresponding one of a plurality of (three) lenses **62** through corresponding one of a plurality of (three) lenses **61**.

Configuration of Container

[0053] As shown in FIGS. 7 to 10, the container **10** includes a container main body **11** for accommodating the sample **200**, and a lid **12** for covering an opening of the container main body **11**. The container main body **11** is a transparent part.

[0054] The container main body **11** is formed of silica glass or borosilicate glass, for example. Also, the lid **12** is a cap including a septum **12a** that can be used for an autosampler. As shown in FIGS. 8 and 9, the lid **12** includes the septum **12a** and an aluminum cap **12b**, and is configured to cover the opening of the container main body **11**. Specifically, the septum **12a** is configured to seal the opening **11a** of the container main body **11** (an end surface of the container main body **11** on an X2-direction side). The septum **12a** includes PTFE (Poly Tetra Fluoro Ethylene), rubber, silicone rubber, polyethylene, or the like, for example. Also, the aluminum cap **12b** is configured to cover the septum **12a**, which seals the container main body **11** and the opening **11a** of the container main body **11**, and is configured to tighten the septum **12a** and the container main body **11**. That is, the aluminum cap **12b** is used as an

aluminum to-be-crimped part. Also, the aluminum cap **12b** is configured to expose a central part of the septum **12a**, as shown in FIGS. **8** and **9**.

[0055] Also, the introduction tube **121a** and the collection tube **122a** pass through the lid of the container **10**. Specifically, the introduction tube **121a** and the collection tube **122a** pass through the central part of the septum **12a** exposed from the aluminum cap **12b**. In this embodiment, the activating gas is supplied from the gas cylinder **104** through the introduction tube **121a**, which passes through the septum **12a**. Also, the gas that includes the generated gas appearing with the deterioration of the sample **200** is sent to the gas chromatograph **103** through the collection tube **122a**, which passes through the septum **12a**. The introduction tube **121a** and the collection tube **122a** are glass capillaries, for example.

[0056] A sample support **13** for supporting the sample **200** in the container **10** is provided (accommodated) in the container **10**. The sample support **13** includes a support plate **13a** for supporting the sample **200** and a magnetic rod **13b**. The magnetic rods **13b** is a rod-like ferromagnetic metal part. The magnetic rod **13b** is formed of a stainless alloy steel containing chromium (Cr) such as SUS430, for example. Also, the magnetic rod **13b** may be plated depending on the activating gas contained in the container **10** and the product appearing with the deterioration of the sample **200** (generated gas).

[0057] Also, as shown in FIG. **10**, the support plate **13a** has a convex U-shaped shape bulging toward a Z1 side (irradiator **60** side) as viewed from an X2-direction side. The support plate **13a** is arranged to surround the three directional side (Z1-direction side, Y1-direction side and Y2-direction side) of the magnetic rod **13b**, and is configured to hold the magnetic rod **13b**.

[0058] In this embodiment, the sample support **13** is configured to support the sample **200** in the container **10**, and to be able to remove the sample from the container **10**. Specifically, the sample support **13** can be removed from the container main body **11** of the container **10** toward the X2-direction side after the lid **12** is removed. Accordingly, since the user can support (place) the sample **200** on the sample support **13** removed from the container **10**, the sample **200** can be easily supported on the sample support **13**.

Configuration of Container Holder

[0059] As shown in FIG. **11**, the container holder **20** includes an accommodation part **21** for accommodating the containers **10**. Also, the container holder **20** includes a cover **22** covering the accommodation part **21** from a top side (Z1-direction side). The accommodation part **21** includes an aluminum block **21a** arranged on the upper surface of the temperature adjuster **30**. Also, thermally insulating parts **21b** and **21c** are arranged on an outer periphery of the aluminum block **21a** to thermally insulate interior space S defined by the accommodation part **21** and the cover **22**. Also, the accommodation part **21** includes a slit plate **21d**, which is arranged above the aluminum block **21a** and has a plurality of (three) slits corresponding to the plurality of (three) containers **10**. Also, the cover **22** includes a quartz plate (silica glass) **22a**, which is a plate-like transparent part for allowing light (UV light) with which the sample is irradiated by the irradiator **60** to enter the accommodation part **21**. Accordingly, the sample in the interior space S defined by

the accommodation part **21** and the cover **22** can be irradiated with the light (UV light) with which the sample is irradiated by the irradiator **60**. Also, the cover **22** includes thermally insulating parts **22b** and **22c** for thermally insulating the interior space S.

[0060] The lens **62** of the irradiator **60** is configured to narrow the irradiation area to correspond to a placement position of the sample **200** in the container **10**. Specifically, the lens narrows the irradiation area of light guided from the light source **50** into an elongated rectangular irradiation area extending in an X direction to agree a shape of the support plate **13a** on which the sample **200** is supported. Accordingly, since the irradiation area is narrowed to agree with the placement position of the sample **200**, the sample **200** can be efficiently irradiated with the light. For example, the sample **200** that has a length of ten and several millimeters in the Y direction and a length of several tens millimeters in the X direction can be placed on the support plate **13a**.

[0061] In this embodiment, the test apparatus main body **110** is configured to apply the accelerated deterioration test to the samples **200** accommodated in the containers **10** by irradiating the sample **200** with the light with which the samples are irradiated through the container main body **11**, which is a translucent part, by the irradiator **60**, with the samples **200** being accommodated in the containers **10**. Specifically, the deterioration of the sample **200** caused by light (UV light) is speeded up (accelerated) by irradiating the sample **200** with the light which is guided from the light source **50** and with which the samples are irradiated through the quartz plate **22a** and the container main body **11** by the irradiator **60**.

[0062] That is, in this embodiment, the test apparatus main body **110** is configured to apply the accelerated deterioration test to the samples **200** accommodated in the containers **10** while adjusting temperature by using the temperature adjuster **30** with the samples **200** being accommodated in the containers **10**. Specifically, the temperature adjuster **30** heats air in the interior space S, which is defined by the accommodation part **21** and the cover **22**, through the aluminum block **21a** of the accommodation part **21** to increase the temperature in the container **10**. Accordingly, the deterioration of the sample **200** caused by temperature (heat) is speeded up (accelerated) by increasing temperature in the container **10**.

[0063] As shown in FIGS. **11** and **12**, the container holder **20** holds the plurality of (three) containers **10** accommodating the measurers **40** together with the samples **200**.

[0064] The measurer **40** includes a temperature sensor **41** accommodated in the container **10**, as shown in FIG. **12**. In addition, the measurer **40** includes a humidity sensor **42**. That is, the measurer **40** can measure temperature and humidity. Here, the temperature sensor **41** and the humidity sensor **42** may be formed integrally with each other. Also, the measurer **40** may include various sensors, such as an illumination sensor, for measuring environments in the container **10**. The three containers **10** for accommodating the samples **200** and the measurers **40** are arranged side by side in the Y direction (front-to-rear direction) in the accommodation part **21**. The samples **200** accommodated in the three containers **10** may be the same as or different from each other.

[0065] Also, the container holder **20** (accommodation part **21**) includes removable container stoppers **81**, **82** and **83** for removably holding the containers **10**. The container stoppers

81, 82 and **83** are configured to be elastically deformed to hold the container **10** from the Y1-direction side, the Y2-direction side, and the X1-direction side. The container stoppers **81, 82** and **83** include plate springs, for example.

[0066] Also, the lid **12** of the container **10** accommodating the measurer **40** is configured to allow a communication cable **43** for transmitting detection signals of the measurers **40** (temperature sensor **41** and humidity sensor **42**) to the outside to pass through the lid. The communication cable **43** is connected to the control board for controlling the accelerated deterioration test apparatus **101** or the PC **102**. Also, the communication cable **43** may be connected to the temperature adjuster **30**, the display (not shown) for displaying temperature in the container **10**, or the like.

[0067] Also, as shown in FIG. 13, the communication cables **43** and the introducer/collectors **120** are arranged to extend from an interior of the container holder **20** to the outside. Also, the container holder **20** includes sample surface adjusters **90** for adjusting inclinations of surfaces **200a** of the samples **200** in the containers **10** to angles that bring so that the surfaces **200a** of the samples **200** perpendicular to a light irradiation direction (Z direction) of the irradiator **60**. Accordingly, since the light can be incident on the surface **200a** of the sample **200** in the container **10** perpendicular to the surface of the sample, the deterioration of the sample **200** can be accurately accelerated. Here, the sample surface adjusters **90** are provided to each of the containers **10**.

[0068] The sample surface adjuster **90** includes a magnet **91** arranged on a side (Z2-direction side) opposite to a side on which the irradiator **60** is arranged with respect to the container **10**. Also, the sample surface adjuster **90** includes a screw **92** for attaching the magnet **91** to the aluminum block **21a** (accommodation part **21**), and a spacer **93** for adjusting a height position (a position in the Z direction) of the magnet **91**.

[0069] Also, the sample support **13** including the magnetic rod **13b**, which is the rod-like ferromagnetic metal part, is configured to be attracted to the magnet **91** to bring the surface **200a** of the sample **200** in the container **10** perpendicular to the direction extending in the irradiation direction (Z direction). Specifically, the support plate **13a** (see FIG. 10), which holds the magnetic rod **13b** and has the U-shape as viewed from the X1-direction side, is adjusted in the container **10** by attracting the magnetic rod **13b** to the magnet **91** so that the U-shape bulges toward the Z1-direction side (toward the irradiator **60**) as viewed from the X1-direction side. Accordingly, even in a case in which the inclination of the surface **200a** of the sample **200** with respect to the irradiation direction is not perpendicular to the irradiation direction of the light with which the sample is irradiated by the irradiator **60** (Z direction) when the sample support **13** holds the container **10**, a magnetic force moves the sample support **13**, which includes the magnetic rod **13b**, to bring the surface **200a** of the sample **200** in the container **10** perpendicular to the direction extending in the irradiation direction (Z direction). As a result, since the surface **200a** of the sample **200** in the container **10** is brought perpendicular to the direction extending in the irradiation direction (Z direction), the light can be accurately incident on the surface **200a** of the sample **200** in the container **10**.

Accelerated Deterioration Test and Analysis Processing of Generated Gases

[0070] The following description describes a processing flow of the accelerated deterioration test and analysis processing of generated gas conducted by the accelerated-deterioration-test analysis system **100** according to this embodiment with reference to FIG. 14.

[0071] In step **901**, the activating gas (gas) is first introduced by control of the PC **102**. In step **901**, the activating gas as the deterioration factor for deteriorating the sample **200** is introduced into the containers **10** by the control of the PC **102** with the samples **200** as the test subject being accommodated in the containers **10**. Accordingly, the accelerated deterioration test for accelerating the deterioration of the samples **200** is activated. In step **901**, heating by the temperature adjuster **30** and irradiation of the samples with ultraviolet rays (UV light) by the irradiator **60** are activated together with the activation of the introduction of the activating gas (gas) (the activation of the accelerated deterioration test). After step **901** is completed, the procedure goes to step **902**. Here, step **901** is an example of a “accelerated deterioration test step” in the claims.

[0072] In step **902**, collection of the generated gases (products) is conducted by the control of the PC **102**. In step **902**, the gases appearing with the deterioration of the samples **200** (generated gases) are collected from the containers **10** by the control of the PC **102** during the accelerated deterioration test. After step **902** is completed, the procedure goes to step **903**. Here, step **902** is an example of a “collection step” in the claims. Also, step **902** may be executed at any time desired by the user or at a predetermined time after the completion of Step **901** (after the activation of the accelerated deterioration test).

[0073] In step **903**, the generated gases (products) are analyzed by the gas chromatograph **103**. In step **903**, the generated gases, which are gases appearing with the deterioration of the samples **200** and are collected from the containers **10** during the accelerated deterioration test, are analyzed during the accelerated deterioration test. In step **903**, the gases appearing with the deterioration of the samples **200** (generated gases) are analyzed by the gas chromatograph **103**. After analysis in step **903** is completed, the procedure goes to step **904**. Here, step **903** is an example of an “analysis step” in the claims.

[0074] In step **904**, analysis results are displayed on the display **103a**. In step **904**, the analysis results of the generated gases analyzed by the gas chromatograph **103** are displayed on the display **103a** by control of a controller (not shown) of the gas chromatograph **103**. Subsequently, if a test time of the accelerated deterioration test reaches the predetermined test time, the procedure goes to step **905**.

[0075] In step **905**, the introduction of the activating gas (gas) is deactivated by the control of the PC **102**. In step **905**, the control of the PC **102** deactivates the introduction of the activating gas (gas) and ends the accelerated deterioration test. In step **905**, heating by the temperature adjuster **30** and irradiation of the samples with ultraviolet rays (UV light) by the irradiator **60** are deactivated together with the deactivation of the introduction of the activating gas (gas).

Advantages of the Embodiment

[0076] In this embodiment, the following advantages are obtained.

[0077] In this embodiment, the accelerated-deterioration-test analysis system 100 (accelerated deterioration test apparatus 101) conducts an accelerated deterioration test by introducing an activating gas (gas) as a deterioration factor into containers 10 accommodating samples 200 as test subjects by using an introducer/collector 120, and collects generated gasses (products) that appear with deterioration of the samples 200 by using the introducer/collector 120. Accordingly, since the activating gas as the deterioration factor is introduced into the containers 10 through the introducer/collector 120 during the accelerated deterioration test, it is possible to maintain conditions for acceleration of deterioration of the samples 200 accelerated by the activating gas. Also, deterioration states of the samples 200 can be acquired during the accelerated deterioration test by analyzing the generated gases collected through the introducer/collector 120 during the accelerated deterioration test. For these reasons, the deterioration states of the samples 200 during the accelerated deterioration test can be acquired while maintaining conditions for acceleration of deterioration of the samples 200. In addition, since the generated gases, which appear with deterioration of the samples 200, are collected during the accelerated deterioration test, it is possible to prevent the generated gases, which appear with deterioration of the samples 200, from affecting the accelerated deterioration test.

[0078] In addition, following additional advantages can be obtained by the accelerated deterioration test apparatus 101 according to the aforementioned embodiment added with configurations discussed below.

[0079] In this embodiment, the containers 10 include container main bodies 11 for accommodating the samples 200, and lids 12 for covering openings of the container main bodies 11. Also, the introducer/collector 120 includes introduction tubes 121a and collection tubes 122a (introduction/collection tubes), which pass through the lids 12 of the containers 10. Accordingly, the activating gas (gas) can be easily introduced through the introduction tubes 121a and the generated gases (products) can be easily collected with the samples 200 being accommodated in the container main bodies 11 (containers 10).

[0080] In this embodiment, to introduce the activating gas as the deterioration factor into the containers 10 and to collect the generated gases as the products, which appear with the deterioration of the samples 200 during the accelerated deterioration test, the introducer/collector 120 is connected to the containers 10 and is connected to a gas chromatograph 103 (analyzer) to be able to send the generated gases, which appear with the deterioration of the samples 200, to the gas chromatograph 103. Accordingly, the generated gases, which appear with the deterioration of the samples 200 caused by the activating gas, can be analyzed during the accelerated deterioration test by the gas chromatograph 103. Consequently, it is possible to speedily acquire deterioration states of the samples 200 during the accelerated deterioration test based on analysis results by using the gas chromatograph 103.

[0081] In this embodiment, the introducer/collector 120 includes an introduction path 121 connecting a gas cylinder 104, which supplies the activating gas (gas) as the deterioration factor into the containers 10, to the containers 10, and a collection path 122 connecting the containers 10 to the gas chromatograph 103. Accordingly, it is possible to separately and independently conduct introduction of the activating gas

(gas) and collection of the generated gases (products) dissimilar to a case in which the path for supplying the activating gas (gas) as the deterioration factor into the containers 10 and the path for connecting the gas chromatograph 103 (analyzer) to the containers 10 are commonly provided. Consequently, it is possible to efficiently introduce the activating gas and efficiently collect the generated gases.

[0082] In this embodiment, the introducer/collector 120 includes a switcher 124 (first switcher) for switching selection of, from the plurality of containers 10, the container 10 from which the product, which appears with the deterioration of the sample 200, is collected. Accordingly, it is possible to separately collect the products, which appear with the deterioration of the samples 200 accommodated in the plurality of containers 10, while conducting the accelerated deterioration test of the sample 200 accommodated in the plurality of containers 10 at the same time.

[0083] Also, in this embodiment, the introducer/collector 120 includes a switcher 123 (second switcher) for switching selection of, from the plurality of containers 10, the container 10 to which the activating gas (gas) as the deterioration factor is introduced. Accordingly, the supply source of the activating gas (gas cylinder 104) can be commonly provided to the plurality of containers 10. Consequently, it is possible to simplify a configuration of the apparatus by reducing the number of supply sources of the activating gas as compared with a case in which supply sources of the activating gas are separately provided to the plurality of containers 10.

[0084] Also, in this embodiment, the accelerated deterioration test apparatus 101 introduces the activating gas (gas) as the deterioration factor into the containers 10 by using the introducer/collector 120 with the samples 200 being accommodated in the containers 10, accelerates the deterioration of the samples 200 while conducting temperature adjustment and irradiation of the samples 200 with light, and collects the products, which appear with the deterioration of the samples 200, by using the introducer/collector 120. Accordingly, it is possible to collect the products, which appear with the deterioration of the samples 200, while accelerating deterioration caused by temperature (heat) of the samples 200 and deterioration of the samples 200 caused by light at the same time in addition to deterioration caused by the activating gas as the deterioration factor.

[0085] In addition, following additional advantages can be obtained by the accelerated-deterioration-test analysis system 100 according to the aforementioned embodiment added with configurations discussed below.

[0086] In this embodiment, the accelerated-deterioration-test analysis system 100 includes a PC 102 (controller) for controlling sending of generated gases (products) appearing with the deterioration of the samples 200 to the gas chromatograph 103 through the introducer/collector 120. Accordingly, since the generated gases, which appear during the accelerated deterioration test, are sent to the gas chromatograph 103 through the introducer/collector 120 by control of the PC 102, it is possible to automatically analyze the generated gases, which appear during the accelerated deterioration test, by using the gas chromatograph 103. Consequently, it is possible to easily analyze the generated gas by using the gas chromatograph 103 even if the accelerated deterioration test is conducted for a long time as

compared with a case in which a user manually controls sending of the generated gases (products) to the gas chromatograph 103.

[0087] Also, in this embodiment, the gas chromatograph 103 analyzes the products, which are collected from the containers 10 during the accelerated deterioration test by using the introducer/collector 120, during the accelerated deterioration test. Accordingly, it is possible to analyze reaction that occurs with the deterioration of the samples 200 during the accelerated deterioration test in real time.

[0088] Also, in this embodiment, the accelerated-deterioration-test analysis system 100 includes a display 103a for displaying analysis results of the products, which appear with the deterioration of the samples 200 and are sent to the gas chromatograph 103 through the introducer/collector 120, analyzed by the gas chromatograph 103. Accordingly, the user can easily grasp the reaction that occurs with the deterioration of the samples 200 due to the accelerated deterioration test by seeing the analysis results displayed on the display 103a.

Modified Embodiments

[0089] Note that the embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present invention is not shown by the above description of the embodiments but by the scope of claims for patent, and all modifications (modified embodiments) within the meaning and scope equivalent to the scope of claims for patent are further included.

[0090] While the example in which the accelerated deterioration test apparatus 101 introduces the activating gas as the deterioration factor into the containers 10 by using the introducer/collector 120 with the sample 200 being accommodated in the container 10, and conducts the accelerated deterioration test has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the accelerated deterioration test apparatus may be configured to introduce a liquid as a deterioration factor into the container by using introducer/collector with the sample being accommodated in the container, and to conduct the accelerated deterioration test. The accelerated deterioration test apparatus may introduce a liquid such as acid liquid, alkaline liquid, salt water, and water into the container, and conduct the accelerated deterioration test. In a case in which an acid liquid is introduced, an acid resistance test of a sample against acids such as acid rain and acidic chemical (accelerated deterioration test) can be conducted. In a case in which an alkaline liquid is introduced, an alkaline resistance test of a sample against alkalis such as alkaline chemicals (accelerated deterioration test) can be conducted. In a case in which salt water is introduced, a resistance test of a sample against seawater and sea breeze can be conducted. In a case in which water is introduced, a water resistance test can be conducted. In the present invention, the accelerated deterioration test apparatus may introduce both a liquid and a gas as deterioration factors and conduct the accelerated deterioration test.

[0091] While the example in which gas chromatograph 103 analyzes the generated gas as the product appearing with the deterioration of the sample 200 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the analyzer for analyzing the product appearing with the deterioration of the sample is not limited to the gas chromatograph. For

example, a liquid or a precipitate may appear as the product with the deterioration of the sample, and be analyzed by a liquid chromatograph or supercritical fluid chromatograph.

[0092] While the example in which the accelerated deterioration test apparatus 101 introduces a gas (activating gas) as a deterioration factor into the containers 10 by using the introducer/collector 120 with the sample 200 being accommodated in the container 10, and accelerates the deterioration of the samples 200 while conducting temperature adjustment and irradiation of the samples 200 with light has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the accelerated deterioration test apparatus may introduce a gas or a liquid as the deterioration factor into the containers by using the introducer/collector with the sample being accommodated in the container, and accelerate the deterioration of the samples 200 while conducting one of temperature adjustment and irradiation of the samples with light. Also, the accelerated deterioration test apparatus may introduce a gas or a liquid into the container by using the introducer/collector only as a deterioration factor with the sample being accommodated in the container to accelerate the deterioration of the sample.

[0093] While the example in which the introducer/collector 120 is connected to the gas chromatograph 103 to be able to send the product appearing with the deterioration of the sample 200 to the gas chromatograph 103 (analyzer) for analyzing the product has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the product collected by using the introducer/collector during the accelerated deterioration test may be collected (stored) in a collection container, and the products collected (stored) in the collection container may be analyzed by an analyzer provided separately from the accelerated deterioration test apparatus during the accelerated deterioration test or after the accelerated deterioration test.

[0094] While the example in which the introducer/collector 120 includes an introduction path 121 connecting a gas cylinder 104, which supplies the activating gas (gas) as the deterioration factor into the containers 10, to the containers 10, and a collection path 122 connecting the containers 10 to the gas chromatograph 103 has been shown in the aforementioned embodiment, the present invention is not limited to this. In this invention, as shown in an accelerated-deterioration-test analysis system 300 (accelerated deterioration test apparatus 301) according to a modified embodiment shown in FIG. 15, an introducer/collector 320 uses a common introduction collection path 321 and common introduction/collection tubes 321a for introduction of the gas or the liquid as the deterioration factor and for collection of products.

[0095] While the example in which the accelerated deterioration test apparatus 101 includes a plurality of (three) containers 10 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the accelerated deterioration test apparatus may include one or two containers, or four or more containers.

[0096] While the example in which the introducer/collector 120 includes a switcher 124 (first switcher) for switching selection of, from the plurality of containers 10, the container 10 from which the generated gas as the product, which appears with the deterioration of the sample 200, is collected

has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, deterioration factor supply sources for supplying gases or liquids into the containers may be separately provided to the plurality of containers.

[0097] While the example in which the introducer/collector 120 includes a switcher 123 (second switcher) for switching selection of, from the plurality of containers 10, the container 10 to which the activating gas (gas) as the deterioration factor is introduced has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, analyzers may be separately provided to the plurality of containers.

[0098] While the example in which the PC 102 (controller) controls selection of the container 10 to which the activating gas (a gas as the deterioration factor) is introduced by using the switcher 123 (second switcher), and selection of the container 10 from which the generated gas (product) is collected by using the switcher 124 (first switcher) has been shown in this embodiment, the present invention is not limited to this. In the present invention, selection of the container to which the gas or a liquid as the deterioration factor switched by the first switcher, and selection of the container from which the product is collected switched by the second switcher may be conducted based on manual operation of the user.

[0099] While the example in which the gas chromatograph 103 (analyzer) of the accelerated-deterioration-test analysis system 100 includes the display 103a for displaying analysis results analyzed by the gas chromatograph 103 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the analysis results analyzed by the analyzer may be displayed on a display provided separately from the analyzer. For example, the display connected to the PC 102 (see FIG. 1) may display the analysis results analyzed by the analyzer.

[0100] While the example in which, as for the containers 10, the temperature adjuster 30 includes a hot plate and increases temperature of air surrounding the containers 10 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the temperature adjuster may be configured to cool air surrounding the containers.

[0101] While the example in which the temperature sensor 41 is accommodated in the container 10 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the accelerated deterioration test apparatus may measure temperature around the container without the temperature sensor accommodated in the container. Also, the temperature of the sample in the container may be measured by acquiring infrared light emitted from the sample by a sensor outside the container.

[0102] While the example in which the irradiator 60 is configured to divert the light emitted from the light source 50, and to irradiate the samples 200, which are accommodated in the plurality of containers 10 held in the container holder 20, with the light has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the irradiator may collectively irradiate the plurality of containers with light emitted from the light source. Also, the accelerated deterioration test apparatus may include shutters (light shields) in the irradiation

tor or the container holder, and be configured to selectively switch each container whether the container is irradiated with the light.

[0103] While the example in which the irradiator 60 is configured including the lens 62 for narrowing the irradiation area to correspond to a placement position of the sample 200 in the container 10 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the sample may be irradiated with light without narrowing the irradiation area of light.

[0104] While the example in which the sample surface adjuster 90 for adjusting an inclination of the surface 200a of the sample 200 with respect to the irradiation direction is provided has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, no sample surface adjuster may be provided, and the user may adjust the inclination of the surface of the sample with respect to the irradiation direction of light by himself or herself.

[0105] While the example in which the sample support 13 including the magnetic rod 13b (rod-like ferromagnetic metal part) is configured to be attracted to the magnet 91 to bring the surface 200a of the sample 200 in the container 10 perpendicular to the direction extending in the irradiation direction (Z direction) has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, a ferromagnetic part may be arranged outside the container 10, and the sample support may include a magnet.

[0106] While the example in which the sample support 13 including the magnetic rod 13b (rod-like ferromagnetic metal part) accommodated in the container 10 is configured to be attracted to the magnet 91 to bring the surface 200a of the sample 200 in the container 10 perpendicular to the direction extending in the irradiation direction (Z direction) has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, a ferromagnetic part may be arranged on an exterior wall of the container (container main body).

[0107] While the example in which the sample support 13 is configured to support the sample 200 in the container 10, and to be able to remove the sample from the container 10 has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the sample support may be fixed in the container.

[0108] Also, while the example in which the procedure of the accelerated deterioration test and the analysis processing of generated gas according to the present invention has been described by using a flow-driven type flowchart in which processes are sequentially performed along a processing flow for sake of illustration in the aforementioned embodiment, the present invention is not limited to this. In the present invention, procedure operations of the accelerated deterioration test and the analysis processing of generated gas may be realized by an event-driven type processing that executes processes in response to occurrence of events. In this case, the procedure operations of the accelerated deterioration test and the analysis processing of generated gas may be realized entirely by such an event-driven type processing or by a combination of such a flow-driven type processing and such an event-driven type processing.

MODES

[0109] The aforementioned exemplary embodiment will be understood as concrete examples of the following modes by those skilled in the art.

Mode Item 1

[0110] An accelerated deterioration test apparatus includes a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample; and an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample.

Mode Item 2

[0111] In the accelerated deterioration test apparatus according to mode item 1, the container includes a container main body for accommodating the sample, and a lid for covering an opening of the main body; and the introducer/collector includes an introduction/collection tube passing through the lid of the container.

Mode Item 3

[0112] In the accelerated deterioration test apparatus according to mode item 1 or 2, the introducer/collector is connected to an analyzer for analyzing the product, which appears with the deterioration of the sample, to be able to send the product to the analyzer.

Mode Item 4

[0113] In the accelerated deterioration test apparatus according to mode item 3, the introducer/collector includes an introduction path connecting a deterioration factor supply source for supplying the gas or the liquid as the deterioration factor into the container, and a collection path connecting the container to the analyzer.

Mode Item 5

[0114] In the accelerated deterioration test apparatus according to any of mode items 1 to 4, a plurality of containers are provided as the container; and the introducer/collector includes a first switcher for switching selection of, from the plurality of containers, the container from which the product, which appears with the deterioration of the sample, is collected.

Mode Item 6

[0115] In the accelerated deterioration test apparatus according to any of mode items 1 to 4, a plurality of containers are provided as the container; and the introducer/collector includes a second switcher for switching selection of, from the plurality of containers, the container into which the gas or the liquid as the deterioration factor is introduced.

Mode Item 7

[0116] In the accelerated deterioration test apparatus according to mode item 3 or 4, to introduce an activating gas as the deterioration factor into the container and to collect a generated gas as the product, which appears with the deterioration of the sample, during the accelerated deterioration

test, the introducer/collector is connected to the container and is connected to a gas chromatograph as the analyzer to be able to send the generated gas, which appears with the deterioration of the sample, to the gas chromatograph.

Mode Item 8

[0117] In the accelerated deterioration test apparatus according to any of mode items 1 to 7, the accelerated deterioration test apparatus is configured to introduce the gas or the liquid as the deterioration factor into the container by using the introducer/collector with the sample being accommodated in the container, to accelerate the deterioration of the sample while conducting at least one of temperature adjustment and irradiation of the sample with light, and to collect the product, which appears with the deterioration of the sample, by using the introducer/collector.

Mode Item 9

[0118] An accelerated-deterioration-test analysis system includes a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample; an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample; and an analyzer for analyzing the product, which is collected by using the introducer/collector.

Mode Item 10

[0119] In the accelerated-deterioration-test analysis system according to mode item 9, a controller for controlling sending of the product, which appears with the deterioration of the sample, to the analyzer through the introducer/collector is further provided.

Mode Item 11

[0120] In the accelerated-deterioration-test analysis system according to mode item 10, the analyzer is configured to analyze the product, which is collected from the container during the accelerated deterioration test by using the introducer/collector, during the accelerated deterioration test.

Mode Item 12

[0121] In the accelerated-deterioration-test analysis system according to mode item 10 or 11, a display for displaying an analysis result of the product, which appears with the deterioration of the sample and is sent to the analyzer through the introducer/collector, analyzed by the analyzer is further provided.

Mode Item 13

[0122] An accelerated deterioration test method includes an accelerated deterioration test step of conducting an accelerated deterioration test for accelerating deterioration of a sample as a test subject by introducing a gas or a liquid as a deterioration factor for deteriorating the sample by using an introducer/collector connected to a container for accommodating the sample to introduce the gas or the liquid as the deterioration factor into the container and to collect a

product that appears with the deterioration of the sample from the container with the sample being accommodated in a container; and a collection step of collecting the product, which appears with deterioration of the sample, from the container by using the introducer/collector during the accelerated deterioration test.

Mode Item 14

[0123] In the accelerated deterioration test method according to mode item 13, an analysis step of analyzing the product, which appears with deterioration of the sample and is collected from the container during the accelerated deterioration test, during the accelerated deterioration test is further provided.

Description of Reference Numerals

- [0124] 10; container
- [0125] 11; container main body
- [0126] 12; lid
- [0127] 100, 300; accelerated-deterioration-test analysis system
- [0128] 101, 301; accelerated deterioration test apparatus
- [0129] 102; PC (controller)
- [0130] 103; gas chromatograph (analyzer)
- [0131] 103a; display
- [0132] 104; gas cylinder (deterioration factor supply source)
- [0133] 120, 320; introducer/collector
- [0134] 121; introduction path
- [0135] 121a; introduction tube (introduction/collection tube)
- [0136] 122; collection path
- [0137] 122a; collection tube (introduction/collection tube)
- [0138] 123; switcher (second switcher)
- [0139] 124; switcher (first switcher)
- [0140] 200; sample
- [0141] 321a; introduction/collection tube

1. An accelerated deterioration test apparatus comprising: a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample; and an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample, wherein the introducer/collector sends the product during the accelerated deterioration test to an analyzer for analyzing the product.
2. The accelerated deterioration test apparatus according to claim 1, wherein the container includes a container main body for accommodating the sample, and a lid for covering an opening of the main body; and the introducer/collector includes an introduction/collection tube passing through the lid of the container.
3. The accelerated deterioration test apparatus according to claim 1, wherein the introducer/collector is connected to

the analyzer for analyzing the product, which appears with the deterioration of the sample, to be able to send the product to the analyzer.

4. The accelerated deterioration test apparatus according to claim 3, wherein the introducer/collector includes an introduction path connecting a deterioration factor supply source for supplying the gas or the liquid as the deterioration factor into the container, and a collection path connecting the container to the analyzer.

5. The accelerated deterioration test apparatus according to claim 1, wherein

a plurality of containers are provided as the container; and the introducer/collector includes a first switcher for switching selection of, from the plurality of containers, the container from which the product, which appears with the deterioration of the sample, is collected.

6. The accelerated deterioration test apparatus according to claim 1, wherein

a plurality of containers are provided as the container; and the introducer/collector includes a second switcher for switching selection of, from the plurality of containers, the container into which the gas or the liquid as the deterioration factor is introduced.

7. The accelerated deterioration test apparatus according to claim 3, wherein to introduce an activating gas as the deterioration factor into the container and to collect a generated gas as the product, which appears with the deterioration of the sample, during the accelerated deterioration test, the introducer/collector is connected to the container and is connected to a gas chromatograph as the analyzer to be able to send the generated gas, which appears with the deterioration of the sample, to the gas chromatograph.

8. The accelerated deterioration test apparatus according to claim 1, wherein the accelerated deterioration test apparatus is configured to introduce the gas or the liquid as the deterioration factor into the container by using the introducer/collector with the sample being accommodated in the container, to accelerate the deterioration of the sample while conducting at least one of temperature adjustment and irradiation of the sample with light, and to collect the product, which appears with the deterioration of the sample, by using the introducer/collector.

9. An accelerated-deterioration-test analysis system comprising:

a container configured to be able to accommodate a sample as a subject of an accelerated deterioration test for accelerating deterioration of the sample as the test subject by using a deterioration factor for deteriorating the sample;

an introducer/collector connected to the container to introduce a gas or a liquid as the deterioration factor into the container and to collect, during the accelerated deterioration test, a product that appears with the deterioration of the sample; and

an analyzer for analyzing the product, which is collected by using the introducer/collector,

wherein the introducer/collector sends the product during the accelerated deterioration test to the analyzer for analyzing the product.

10. The accelerated-deterioration-test analysis system according to claim 9 further comprising a controller for controlling sending of the product, which appears with the deterioration of the sample, to the analyzer through the introducer/collector.

11. The accelerated-deterioration-test analysis system according to claim **10**, wherein the analyzer is configured to analyze the product, which is collected from the container during the accelerated deterioration test by using the introducer/collector, during the accelerated deterioration test.

12. The accelerated-deterioration-test analysis system according to claim **10** further comprising a display for displaying an analysis result of the product, which appears with the deterioration of the sample and is sent to the analyzer through the introducer/collector, analyzed by the analyzer.

13. An accelerated deterioration test method comprising:
an accelerated deterioration test step of conducting an accelerated deterioration test for accelerating deterioration of a sample as a test subject by introducing a gas or a liquid as a deterioration factor for deteriorating the sample by using an introducer/collector connected to a container for accommodating the sample to introduce

the gas or the liquid as the deterioration factor into the container and to collect a product that appears with the deterioration of the sample from the container with the sample being accommodated in a container; and
a collection step of collecting the product, which appears with deterioration of the sample, from the container by using the introducer/collector during the accelerated deterioration test,

wherein the collection step includes sending the product during the accelerated deterioration test to an analyzer for analyzing the product by using the introducer/collector.

14. The accelerated deterioration test method according to claim **13** further comprising an analysis step of analyzing the product, which appears with deterioration of the sample and is collected from the container during the accelerated deterioration test, during the accelerated deterioration test.

* * * * *